CSA4704-DEEP LEARNING

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Experiment 9: Multilayer Perceptron Implementation
Problem Statement: A telecom operator wants to predict which customers will cancel their subscription next month. A customer-churn dataset with tenure, monthly charges, contract type, complaints and data usage, with a Churn label. The objective is to build, train and evaluate an MLP classifier in Keras. Use TensorFlow, Keras, NumPy, Matplotlib.
Tasks:
1. Pre-process the data: encode categorical fields and scale numeric fields.
2. Build an MLP with two or three hidden layers using ReLU and a sigmoid output.
3. Train with binary cross-entropy and Adam; plot training and validation curves.
4. Report accuracy, precision, recall, F1 and the confusion matrix.
5. Vary the hidden-layer width and note its effect. import numpy as np

FORMULA:

1. Min-Max Feature Scaling

$X_scaled = (X - X_min) / (X_max - X_min)$

2. One-Hot Encoding

Categorical values are converted into binary vectors.

Example: Contract = Monthly → [1, 0, 0]

3. ReLU Activation Function

$f(x) = \max(0, x)$

4. Sigmoid Activation Function

$\sigma(x) = 1 / (1 + e^{-x})$

5. Binary Cross-Entropy Loss Function

$L = -(1/N) \Sigma [y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})]$

6. Adam Optimizer

$\theta_{t+1} = \theta_t - \eta \, \hat{m}_t / (\sqrt{\hat{v}_t} + \varepsilon)$

7. Accuracy

Accuracy = (TP + TN) / (TP + TN + FP + FN) × 100%

8. Precision

Precision = TP / (TP + FP)

9. Recall

Recall = TP / (TP + FN)

10. F1-Score

F1-Score = 2 × (Precision × Recall) / (Precision + Recall)

11. Confusion Matrix

	Predicted Churn	**Predicted No Churn**
Actual Churn	True Positive (TP)	False Negative (FN)
Actual No Churn	False Positive (FP)	True Negative (TN)

12. Training and Validation Loss

Training Loss = Loss calculated on training data

Validation Loss = Loss calculated on validation data

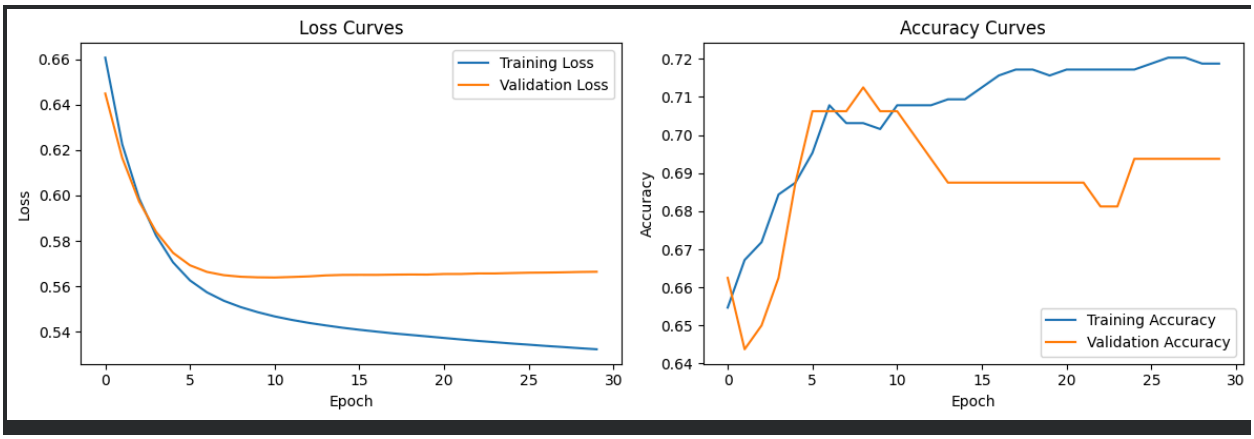
CODE:

```
import pandas as pd
import matplotlib.pyplot as plt
import tensorflow as tf

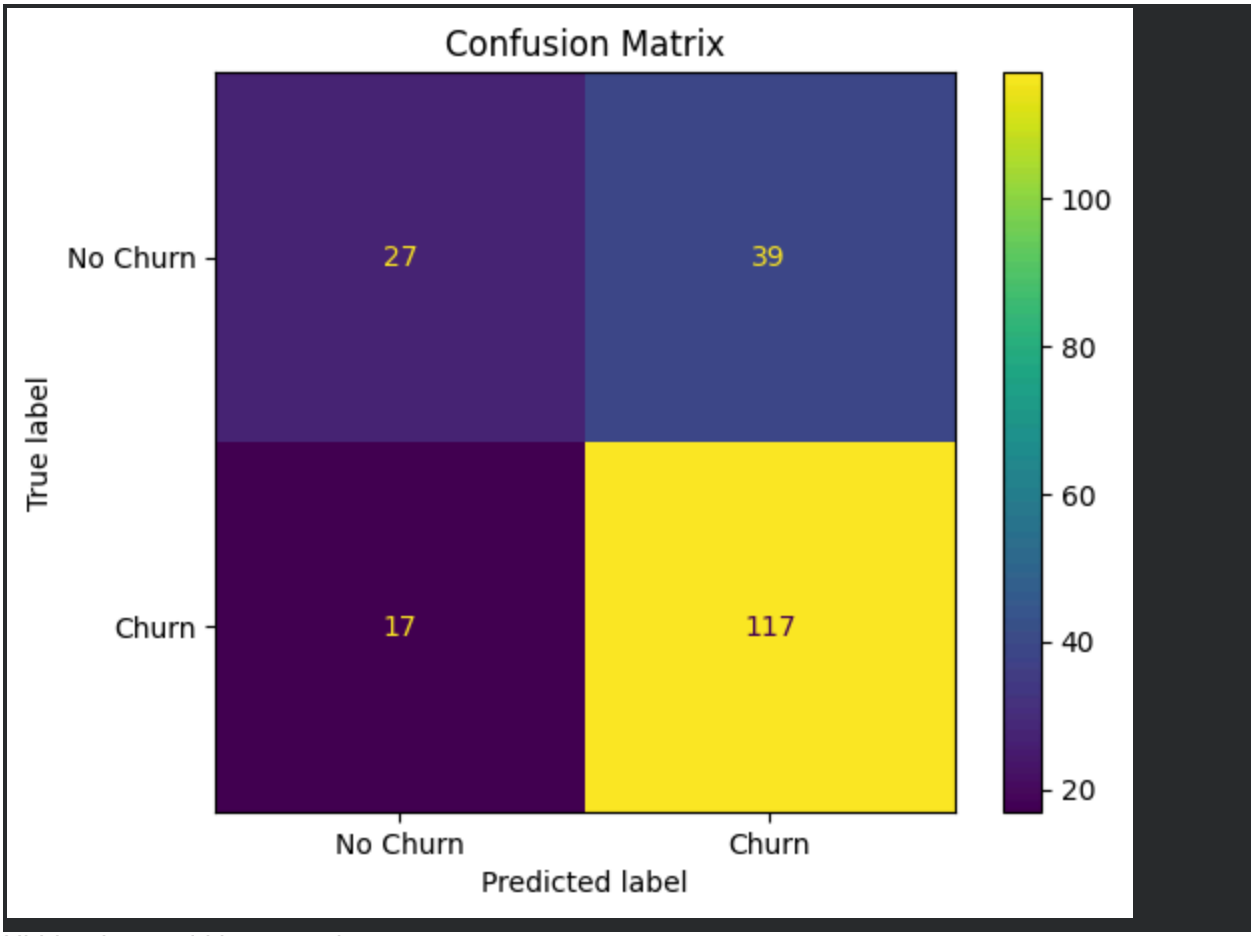
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import (
    accuracy_score, precision_score, recall_score,
    f1_score, confusion_matrix, ConfusionMatrixDisplay
…
```

```
results_df = pd.DataFrame(
    results,
    columns=["Hidden Width", "Accuracy", "Precision", "Recall", "F1"]
)
print("\nHidden-layer width comparison:")
print(results_df.round(3).to_string(index=False))
```

OUTPUT:



Accuracy : 0.72
Precision: 0.75
Recall : 0.873
F1 Score : 0.807



Hidden-layer width comparison:

Hidden Width	Accuracy	Precision	Recall	F1
16	0.700	0.750	0.828	0.787
32	0.695	0.739	0.843	0.787
64	0.695	0.739	0.843	0.787